

Multiple Choice (10 marks)

1. By what factor would the heat transfer by radiation between the walls of a thermos bottle be increased if the reflectance of each wall were reduced from 0.9 to 0.3?
 - (a) 11.3
 - (b) 7.0
 - (c) 13.7
 - (d) 6.5
 - (e) 12.5
2. Compared with the sound you hear from the siren of a stationary fire engine the sound you hear when it approaches you has an increased
 - (a) volume
 - (b) timbre
 - (c) speed
 - (d) duration
 - (e) wavelength
3. How many calories are developed in 1.0 min in an electric heater which draws 5.0 amp when connected to a 110-volt line?
 - (a) 5.0×10^3 cal
 - (b) 15.3×10^3 cal
 - (c) 10.5×10^3 cal
 - (d) 12.7×10^3 cal
 - (e) 7.9×10^3 cal
4. Calculate the mass of the electron by combining the results of Millikan's determination of the electron charge q and J.J. Thomson's measurement of q/m .
 - (a) $6.7 \times 10^{-17}kg$
 - (a) $2.2 \times 10^{-22}kg$
 - (a) $3.0 \times 10^{-32}kg$
 - (a) $5.5 \times 10^{-31}kg$
 - (a) 1.76×10^{11} kg

5. When an element ejects an alpha particle, the mass number of the resulting element
- (a) remains the same
 - (b) increases by 4
 - (c) reduces by 3
 - (d) increases by 3
 - (e) increases by 1

True / False (10 marks)

Answer True or False

- 6. True or False: The force acting on the particle is 3.0 N, as it corresponds to the speed of 3.0 m/s without considering energy changes.
- 7. True or False: The initial charge of the capacitor was 16 q, given it loses half its charge every second for five seconds.
- 8. True or False: The magnification of the image formed by a concave mirror in this scenario is 4 times the size of the object.
- 9. True or False: A force of 0.20 newton acting on a mass of 100 grams results in an acceleration of 2.0 m/sec^2 according to Newton's second law.
- 10. True or False: Huygens' principle describes light propagation as a wavefront, where every point on a wavefront serves as a source of secondary waves.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the trajectory of a ballistic rocket launched at the equator at an angle of 60° with an initial speed of 6 km/sec . Calculate the great circle distance covered around the Earth and analyze the implications of your findings.
- 11. Discuss the dynamics of a head-on collision between a particle of mass m moving at velocity u_1 and a stationary particle of mass $2m$. Given a coefficient of restitution that maximizes kinetic energy loss, derive the final velocity v_1 of the first particle after the collision and analyze the implications of your findings.

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